

WILLIAMS RACING PERFORMANCE ANALYSIS (2015-2019)

HOW CAN PAST RESULTS INFORM CURRENT STRATEGY AT WILLIAMS RACING?

FRANK DONG – SEP 2025

EXECUTIVE SUMMARY

This project applied exploratory data analytics to Williams Racing's performance from 2015 to 2019 to identify underperformance trends, warning signs, and development opportunities for improved race weekend outcomes and consistency.

This project combined exploratory analysis, visualisation, and hypothesis testing of Williams' performance from 2015-2019. Using structured datasets and API-sourced telemetry, three key strategic questions were investigated across race outcomes, qualifying deficits, and driver consistency.

Based on the above process, the following conclusions are identified:

1. Williams showed the greatest volatility in race position changes compared to midfield rivals.
2. Balanced sectors, not high-downforce ones, revealed the largest qualifying deficits.
3. Rookies proved less consistent than veterans, showing race consistency is linked to tenure.

The following recommendations are provided:

1. **Adopt live sector tagging and deficit tracking** to pinpoint performance losses in real time and support faster setup decisions.
2. **Prioritise setup experimentation on balanced circuits** to stabilise performance on short straights, medium speed turns and mixed layouts.
3. **Focus on rookie consistency training** to accelerate the development curve and reliability of all new Williams drivers.

STRATEGIC DESIGN

This project adopts a rigorous, business-first approach, ensuring analysis is anchored to the strategic needs of Williams Racing. Every action taken throughout follows the central goal of turning curiosity about on-track performance into clear, quantitative insights that support confident decision-making.

STRATEGIC QUESTIONS (SQ)

Brainstorming began with high-impact, open-ended strategic questions – those that matter most to team strategy and results.

The selected SQs are:

1. Which circuits saw Williams lose the most positions from race start to finish during the 2015–2019 F1 seasons, and what track characteristics explain these losses?
2. In qualifying sessions, where did Williams lose the most time relative to the fastest midfield team, and were these gaps more pronounced in technical or power-focused sectors?
3. Were rookies less consistent than experienced teammates, and what interventions can improve consistency?

KPIS

Each SQ was translated into a **key performance indicator (KPI)** – a metric that could be measured using actual racing data.

For instance,

1. Race position volatility was quantified through **average positions gained or lost**.
2. **Qualifying deficits** were broken down into sector time gaps, **relative to the fastest midfield team** of Haas, Renault, or Racing Point / Force India.
3. Driver consistency was measured by **lap time standard deviation**.

Developing specific KPIs ensures answers are evidence-driven and transparent to all stakeholders.

HYPOTHESES

We developed an **assumption or testable hypothesis** for each KPI. This step bridges the gap from curiosity to targeted analysis, allowing us to determine not just what happened, but also why.

We tested if:

1. More positions were lost at technical tracks compared to the midfield rival average.
2. More time was lost in the technical sectors in qualifying
3. Rookies had more variable lap times than veterans.

Connecting Strategy to Analysis

#	Strategic Question	KPI	Hypothesis
1	Where and why did Williams lose the most race positions?	Avg. race positions gained/lost	More positions were lost at technical tracks compared to midfield rival average
2	Time lost in qualifying: technical or power sectors? *	Sector time gap to fastest midfield rival	More time was lost in technical sectors
3	Were rookies less consistent than experienced teammates?	Lap time standard deviation, per race	Rookies had more variable lap times

* Limited to 2018 and 2019 race data

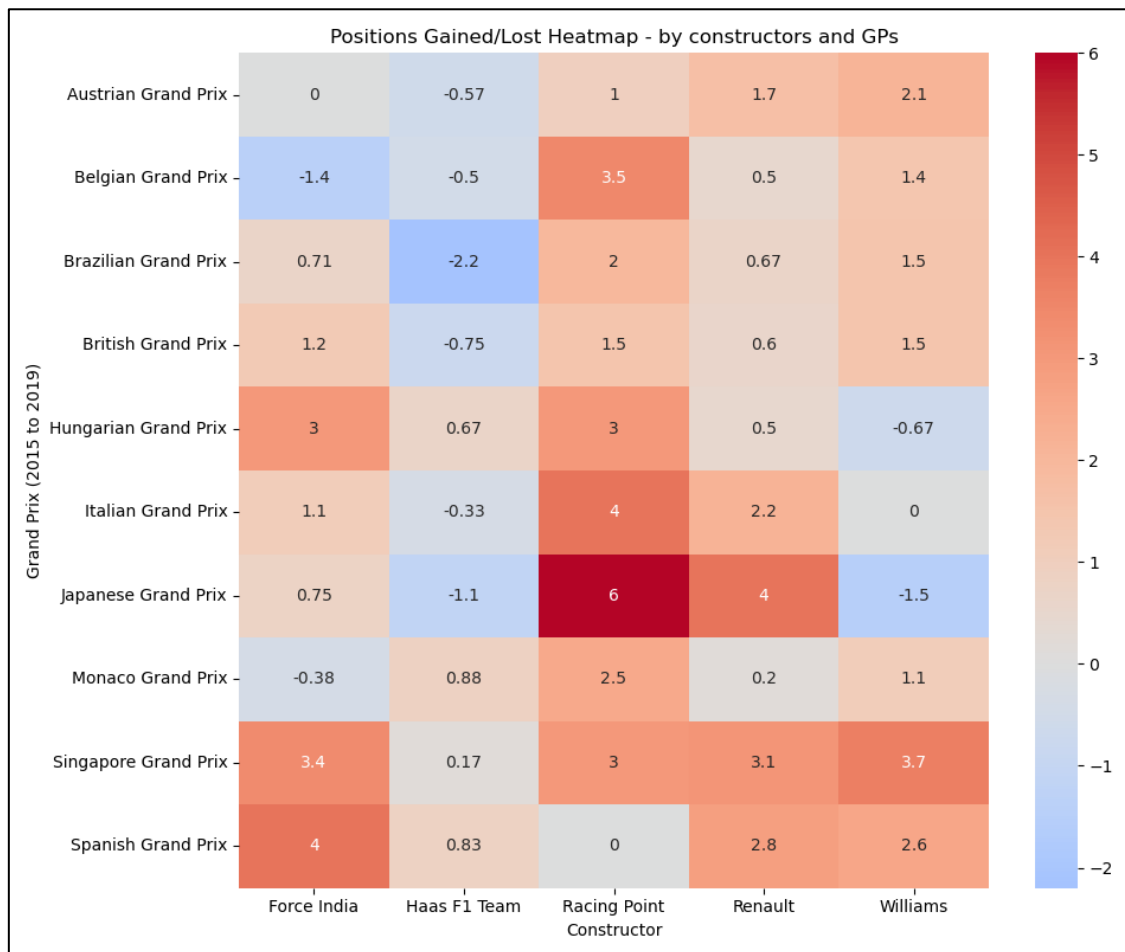
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Connecting SQs to KPIs to hypotheses ensured all elements of the analysis were traceable and relevant – from the initial business need to final recommendations.

This logical structure supports the confidence needed for high-stakes race decisions, with all conclusions grounded in data, not anecdotes.

STRATEGIC QUESTION 1 – GRID- TO-FINISH PERFORMANCE

PLOT 1: POSITION CHANGE HEATMAP



This heatmap compares Williams’ race position changes to those of key midfield rivals – Haas, Renault, Force India and Racing Point – across ten Grand Prix events from 2015 to 2019.

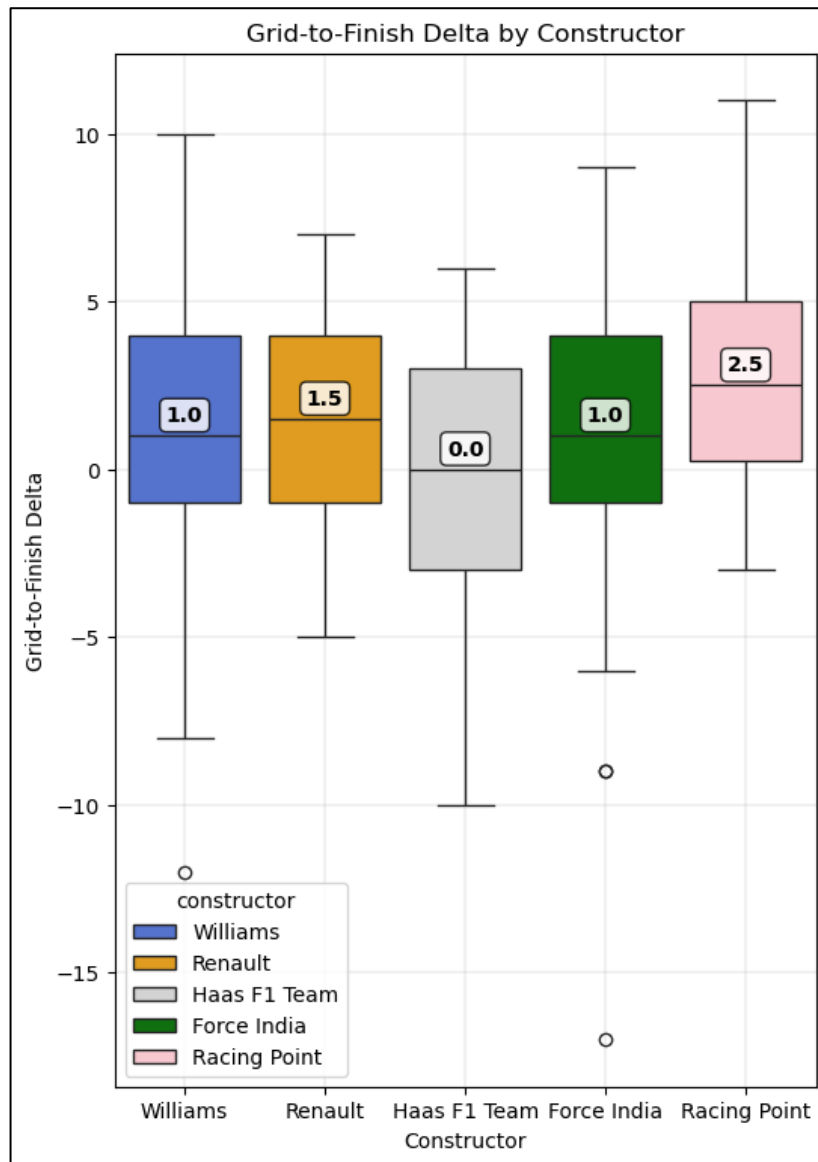
Each cell displays the average positions gained or lost by a team, race by race. Red cells highlight races and teams where position gains were strongest, while blue cells mark races where positions were typically lost.

Racing Point’s deep red in the Japanese GP column, for example, signifies exceptional gains at a track that tends to challenge the midfield.

For Williams, noticeable blue regions – especially at the Japanese and Hungarian GPs – signal a pattern of position loss on circuits known for technical complexity. This supports the hypothesis that high-downforce tracks represent a particular challenge for Williams.

For decision-makers, the heatmap immediately surfaces where Williams’ race outcomes diverge from rivals, signalling tracks that demand focused setup or strategic adjustment.

PLOT 2: POSITION CHANGE DELTA BOXPLOT – BY TEAM



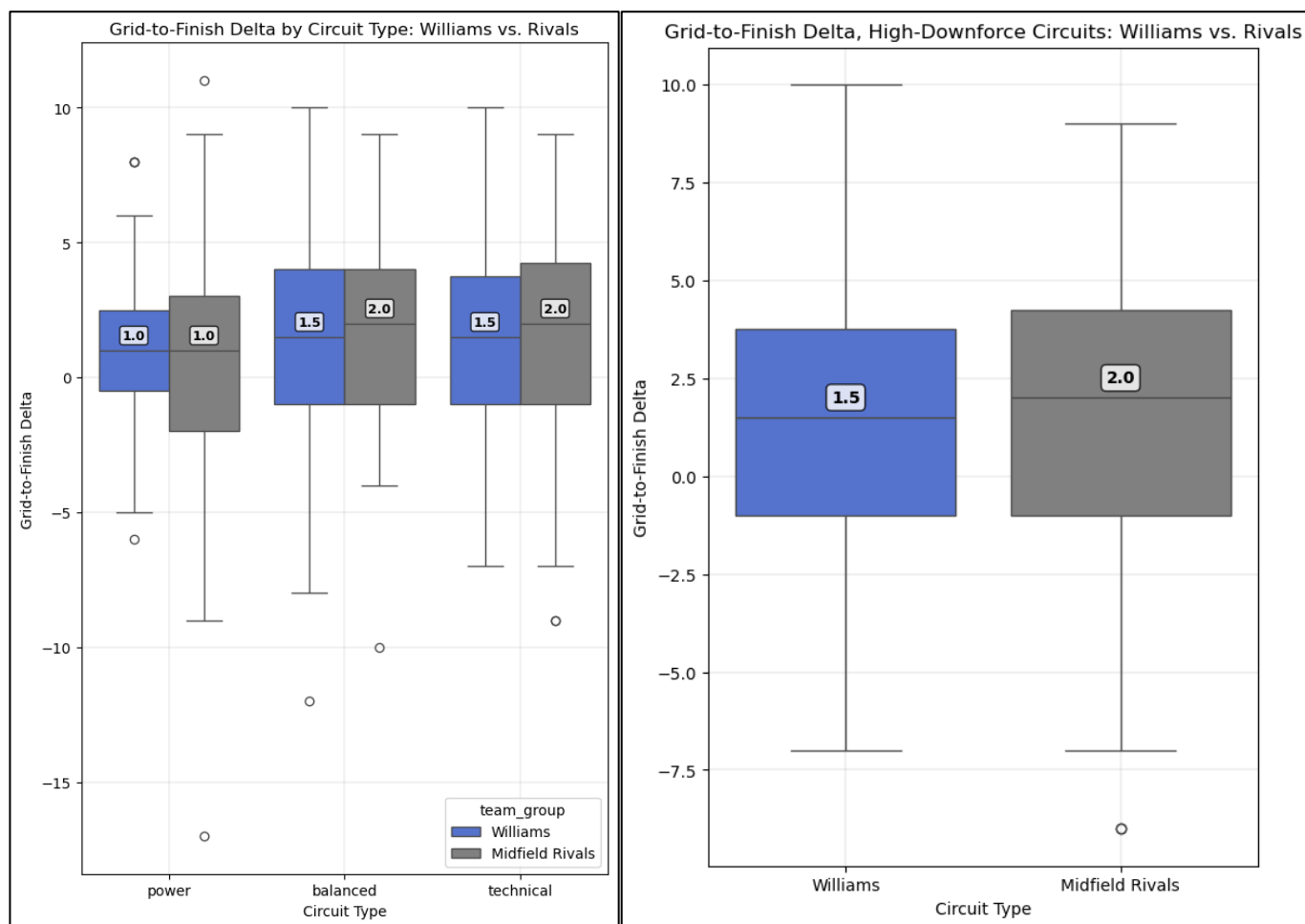
Boxplots are a way to visualise measures of central tendency and spread together. Each colour column represents a team, summarising where their race positions tend to fall, across ten races over five years.

The box displays typical results (interquartile range), while the line inside the box represents the median positions gained or lost. Lines, or whiskers, reveal the maximum positions a team can gain or lose in normal circumstances. Circles represent outliers – rare, standout races which could be good or bad for the team.

Haas shows the greatest variability in 50% of their race outcomes, with the tallest IQR box, while Racing Point shows an improvement from Force India's results.

For Williams, its positive median (+1.0 places) means it tends to gain positions on average. However, its race results swing widely, as it has the longest whiskers compared to all midfield rivals.

PLOT 3 (A) & (B) – POSITION CHANGE DELTA BOXPLOTS – WILLIAMS VS. RIVALS



These two boxplots provide a comparative view of Williams’ race position changes (“grid-to-finish delta”) against other midfield teams, grouped as “Midfield Rivals”.

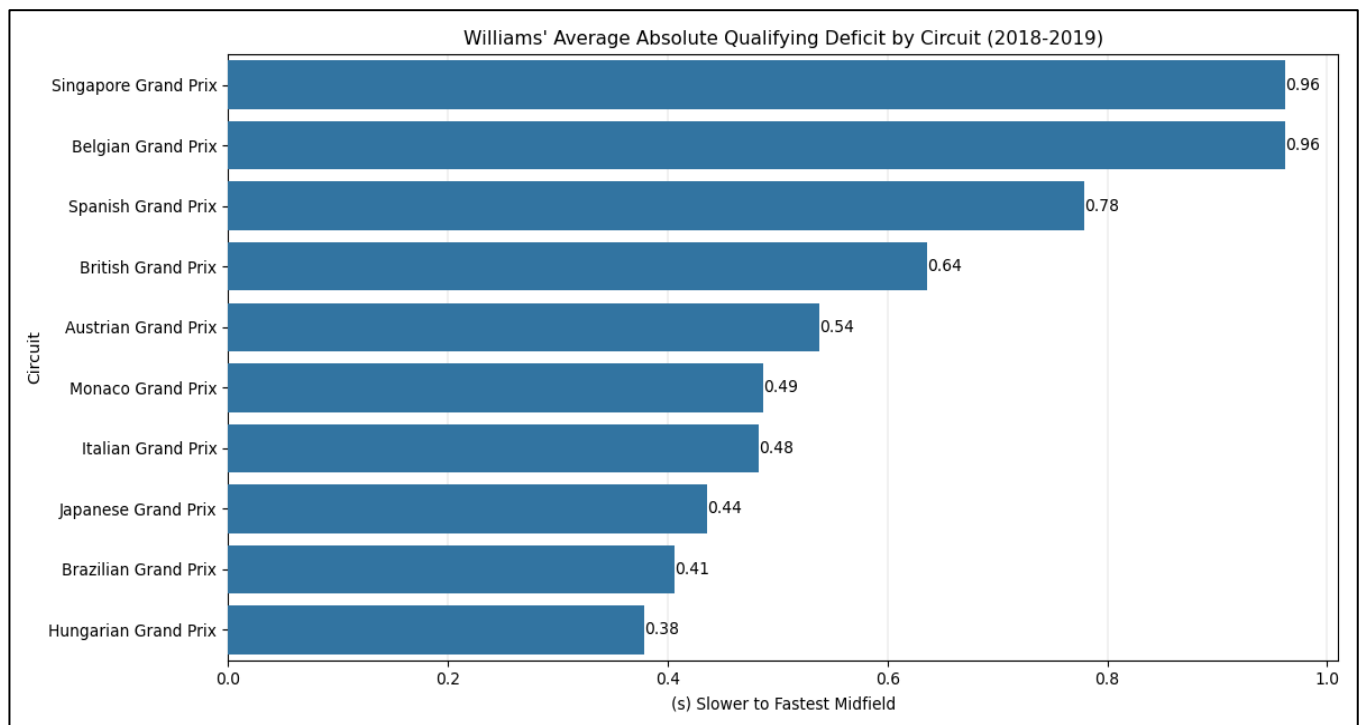
By clustering the data by circuit type – power, balanced, and technical – these visuals allow direct benchmarking of Williams’ performance under a variety of track conditions.

- **Plot 3 (A) (left)** summarises all race sessions across power (e.g. Monza), technical (e.g. Monaco, Hungary) and balanced (e.g. Silverstone, Interlagos) circuits, displaying the distribution and median of position changes for Williams and its competitors.
- **Plot 3 (B) (right)** zooms in specifically on high-downforce, technical circuits, providing a focused comparison where the perception of Williams’ weakness is strongest.

The visuals reveal that Williams’ median race-day results – and their variability – are remarkably like those of the broader midfield group across all circuit types. Notably, on technical tracks, where Williams is often thought to struggle most, both the range and central tendency of performance match those of rivals.

This challenges our initial assumptions: if position losses are not disproportionately higher for Williams at technical circuits, then other factors, like raw pace, setups, or driver abilities, may play a bigger role in shaping race outcomes.

PLOT 4 – QUALIFYING DEFICITS BAR CHART



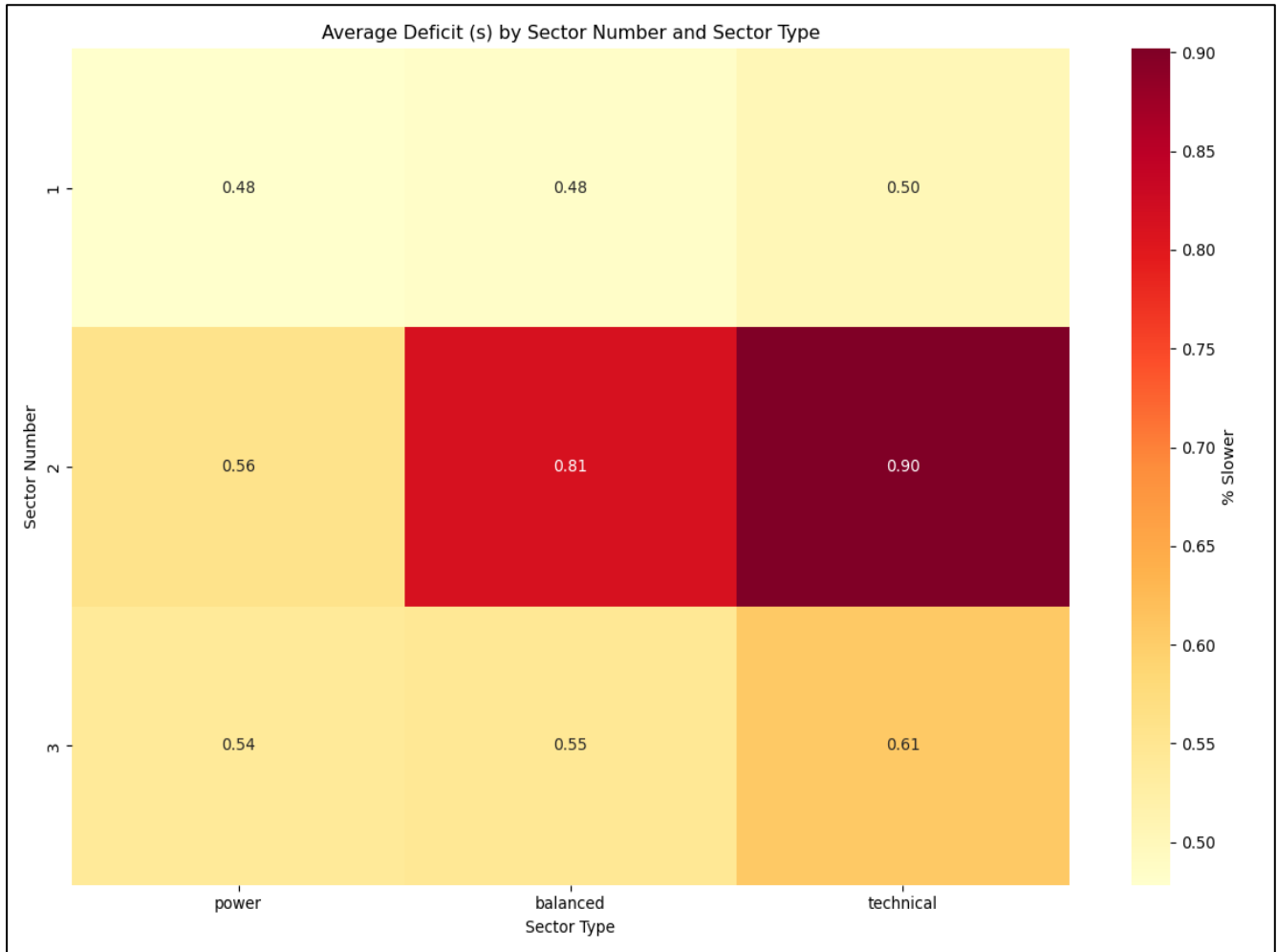
This bar chart quantifies Williams' average qualifying lap time deficit to the fast midfield competitor – Haas, Renault, Force India or Racing Point – at each of the ten selected circuits for 2018 and 2019 (see [Data: Limitations](#)). Sourced from Fast F1 telemetry data, the chart highlights where qualifying performances are the largest.

Key findings reveal that significant time losses appear at circuits like Singapore, Spa-Francorchamps (Belgium) and Spain.

Notably, these gaps do not always correspond with conventional circuit types – large deficits are found on tracks with both technical and balanced features. For example, Spa combines high-speed straights and complex, low-speed sectors, making it challenging to classify and signalling that a broad mix of track elements can contribute to qualifying struggles.

For Williams' decision-makers, this visual directly identifies which venues expose the team's greatest qualifying shortcomings. It underlines the need for sector-level analysis, rather than assumptions based solely on circuit type. Breaking down qualifying gaps in this manner sets the stage for deeper insights – pinpointing whether time is lost in technical or power-heavy sectors – which is key for informed strategic decision-making.

PLOT 5 – SECTOR DEFICITS HEATMAP



This heatmap breaks down Williams’ qualifying performances into sectors, segmented by both sector types – power, balanced, and technical – and by their sequence within each circuit.

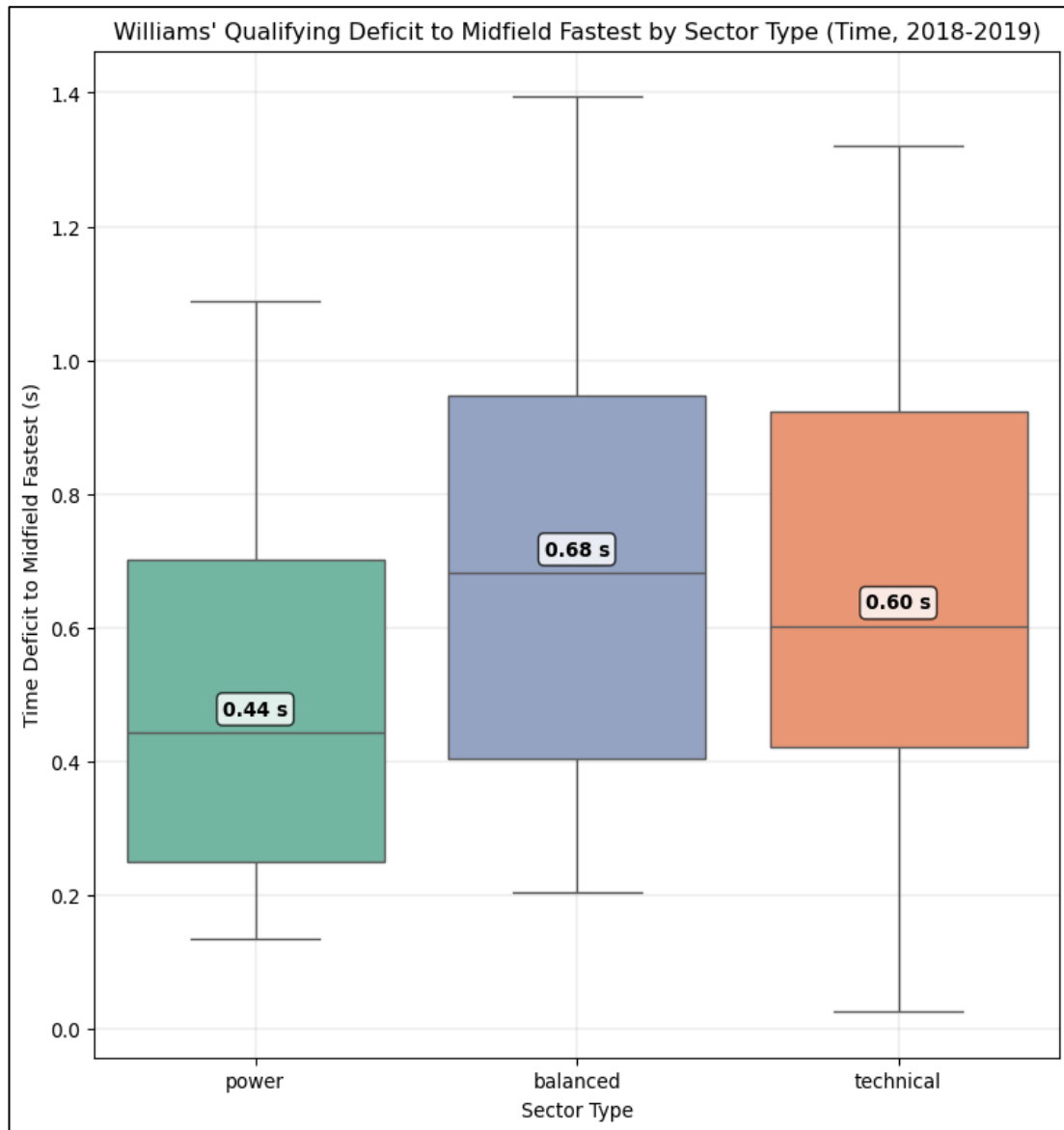
The x-axis organises sector types, while the y-axis orders sector number, revealing where gaps emerge relative to the fastest team.

Deeper red shades indicate larger deficits. The standout finding is Technical Sector 2, with Williams losing almost a second on average, while Balanced S2 and Technical S3 are also areas with substantial deficits.

This heatmap pinpoints vulnerabilities with precision, confirming that Williams faces challenges in technical sectors – especially as the lap progresses into Sectors 2 and 3. This focuses attention on engineering setups and execution improvements in these specific track segments, rather than on power-heavy areas.

Overall, this chart supports our assumption that technical sectors are a critical limitation in Williams’ qualifying pace, guiding future resource allocation and setup refinements.

PLOT 6 – SECTOR DEFICIT BOXPLOT, BY SECTOR TYPE



This boxplot compares Williams' averaging qualifying deficits to the fastest midfield team, broken down by sector type: power, balanced, and technical.

- Williams posts its smallest qualifying gaps in power sectors, losing 0.44 s per lap.
- Technical sectors present a greater challenge, with higher average deficits (~0.6 s) and the most session-to-session variability (whisker range).
- Surprisingly, balanced sectors, which challenge setups on mixed layouts, show both the largest average deficit (~0.7 s) and the widest spread of results (interquartile range), suggesting they may be especially unpredictable for the team.

This chart helps Williams' strategists and engineers pinpoint where qualifying pace is most reliably achieved, and where it consistently slips. Notably, it guides our attention towards balanced and technical sectors as high-value targets for setup refinement and resource allocation, rather than power sectors where performance is already stable. Strategic Question 3 – Driver Consistency

STRATEGIC QUESTION 3 – DRIVER CONSISTENCY

PLOT 7 – LAP TIME CONSISTENCY STRIP PLOT



This strip plot visualises the consistency of each Williams driver by showing the standard deviation of their lap times in individual races between 2015 and 2019.

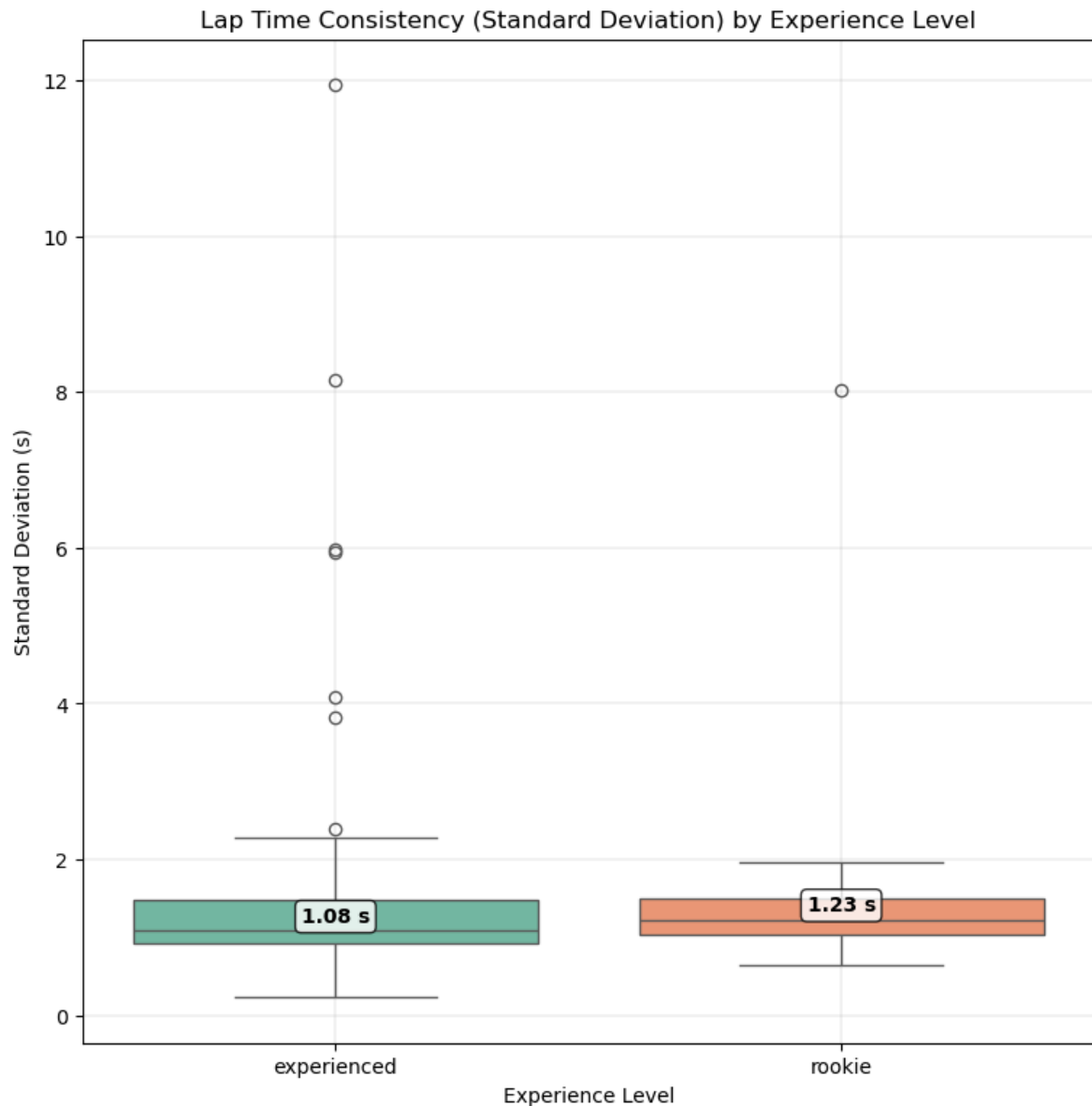
The x-axis separates experienced drivers from rookies, while each driver's results are shown as individual points, colour-coded for clarity.

- Due to more seasons spent with Williams, drivers like Massa and Bottas have more race entries, resulting in denser clusters of points. Kubica, Sirotkin and Russell appear less frequently, reflecting shorter tenures.
- The plot allows us to spot both clusters – where typical lap time consistency falls – and outliers, representing races with unusually high variability in lap times.

For Williams, this plot provides a comprehensive landscape of where driver consistency sits – not just between rookies and veterans but across all individual performances. Clusters help identify reliability, while outliers flag events worth investigating, such as pit stop strategy or setup changes.

Setting out every race result for each driver lays the groundwork for comparing overall consistency between experience levels, clarifying whether rookies truly exhibit greater inconsistency, or if performance is shaped more by context and team conditions.

PLOT 8 – LAP TIME CONSISTENCY BOX PLOT



This boxplot contrasts the lap time consistency of Williams’ rookie drivers versus experienced teammates from 2015-2019, using the standard deviation of race lap times as the measure. It visualises both the spread and the central tendency for each group.

- Results appear to show that rookies are less consistent compared to their experienced counterparts – 1.08 s median lap time std. dev. for veterans versus 1.23 s for rookies.
- While experienced drivers display a lower median standard deviation (implying more consistent laps on average), overall variability is greater. Outlier races appear where consistency broke down.

From this plot, we could conclude that rookies are less consistent than experienced drivers, aligning with our assumptions and Hypothesis 3. An alternative (H_1) hypothesis test result (see [Hypothesis Testing](#)) with both data samples confirms our insights and prioritises immediate attention towards rookie race consistency development.

HYPOTHESIS TESTING

To ensure strategic insights are statistically robust, hypothesis tests were conducted on three core performance questions for Williams Racing. For clarity and completeness, each test includes the business context, formal hypotheses, results summary, and actionable conclusions.

TEST 1: RACE POSITION CHANGE – HIGH DOWNFORCE CIRCUITS

Strategic Question: Did Williams lose significantly more positions on high-downforce (“technical”) tracks versus midfield rivals from 2015–2019?

Hypotheses:

- **Null (H_0):** No difference in grid-to-finish delta between Williams and rivals.
- **Alternative (H_1):** Williams has a worse grid-to-finish delta on high-downforce circuits.

Methods: Welch’s t-test (for group mean comparison) and Mann-Whitney U (non-parametric) to address sample size and distribution issues.

Results:

Test	Test Statistic	p-value	Conclusion (95% confidence)
Welch’s t-test	0.0030	0.1452	Fail to reject null (H_0)
Mann-Whitney U	725.000	0.2299	Fail to reject null (H_0)

Interpretation: No statistically significant evidence Williams lost more grid positions on technical tracks than midfield rivals, regardless of rival outlier handling or test type.

Stakeholder Value: Williams’ race results on technical tracks are not uniquely worse than other midfield teams; setup decisions and race strategy may be as critical as raw car performance for Sunday outcomes.

TEST 2: QUALIFYING SECTOR DEFICITS – TECHNICAL VS. POWER SECTORS

Strategic Question: Does Williams suffer significantly bigger qualifying time losses in technical sectors compared to power sectors?

Hypotheses:

- **Null (H_0):** No significant difference in qualifying sector deficits (technical vs. power).
- **Alternative (H_1):** Technical sector deficits are significantly worse.

Methods: Mann-Whitney U test used to account for small, skewed samples.

Results:

Test	Test Statistic	p-value	Conclusion (95% confidence)
Absolute Deficit (sec slower)	204.000	0.1382	Fail to reject null (H_0)
Relative Deficit (% slower)	181.000	0.2570	Fail to reject null (H_0)

Interpretation: No significant difference found. Current qualifying gaps in technical sectors cannot be conclusively said to be larger than in power sectors.

Stakeholder Value: Qualifying challenges span multiple sector types, so setup and development should address broad car performance; future analysis with larger samples is recommended before targeting technical sectors for improvement.

TEST 3: DRIVER CONSISTENCY – ROOKIES VS. EXPERIENCED

Strategic Question: Were rookie drivers less consistent in race laps, as measured by lap time variance, than veterans?

Hypotheses:

- **Null (H_0):** No significant difference in lap time standard deviation (rookies vs. experienced).
- **Alternative (H_1):** Rookies show higher lap time standard deviation

Methods: Mann-Whitney U for robustness to sample size and distribution shape.

Results:

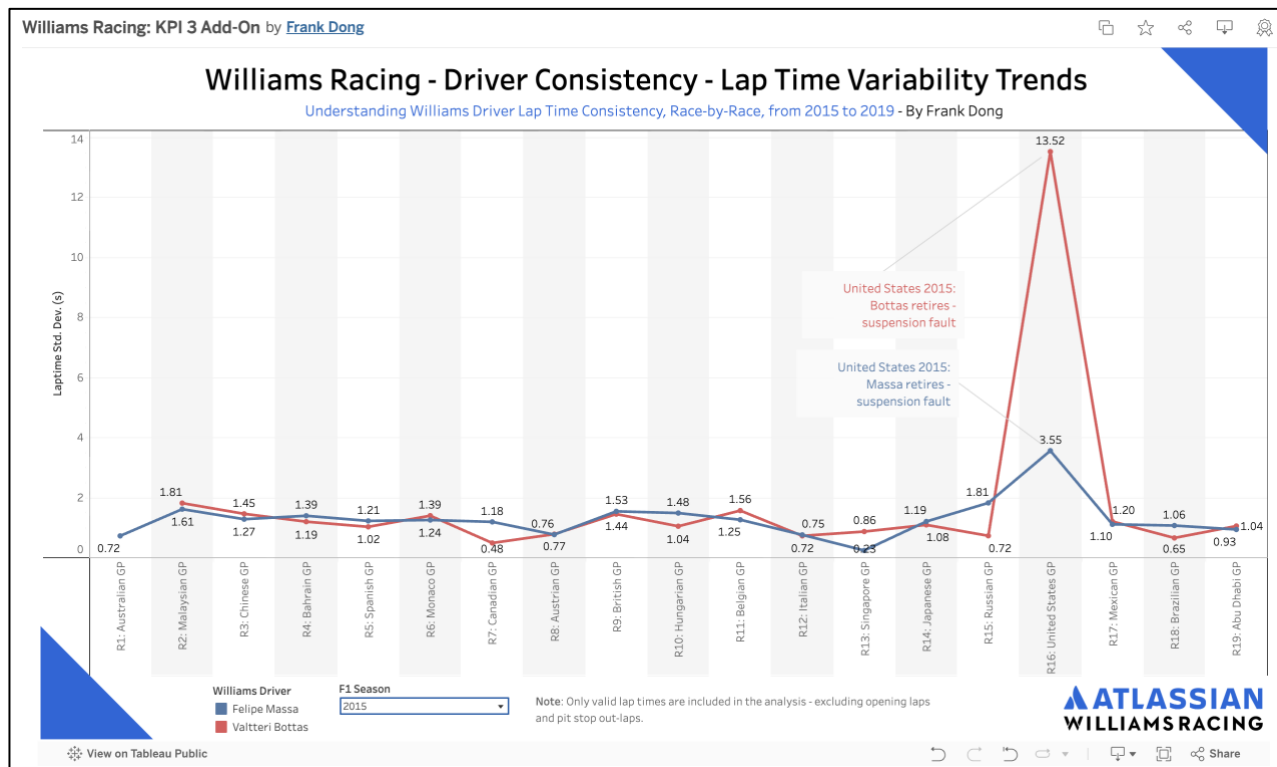
Test	Test Statistic	p-value	Conclusion (95% confidence)
Mann-Whitney U	2632.000	0.0372	Reject null (H_1)

Interpretation: Rookie Williams drivers exhibit greater lap time variance than their more experienced teammates.

Stakeholder Value: Driver consistency and race experience are linked in this sample window – race reliability comes with tenure. Interventions should focus on rookie race consistency training.

ADD-ON INTERACTIVE DASHBOARD

A central deliverable of this project is the [Interactive Tableau Dashboard](#), which offers William Racing stakeholders a dynamic tool to explore and understand the data beyond static reports or slide decks.



A key strength of the Tableau dashboard is its interactivity, making advanced F1 analysis accessible and customisable for all stakeholders. The primary filter, **F1 Season**, allows users to toggle between any season from 2015 to 2019, enabling direct, race-by-race tracking of KPI 3, 'Lap Time Consistency', as well as all other core metrics, providing seasonal context to variability and trends.

Stakeholders can:

- **Select any season** to view changes in driver consistency and identify years or events with notable spikes in lap time variability.
- **Toggle by driver or GP** to compare performances across the whole team or focus in on individual races and performances for deeper investigation.
- **Explore all races in context:** The dashboard includes every GP in the dataset, providing comprehensive coverage and supporting year-over-year performance comparisons.

This flexible filtering helps strategists, engineers, and management move quickly from a "big picture" scan of multiple years to a focused diagnosis of specific events or drivers, empowering data-driven inquiry, scenario planning, and rapid investigation of performance questions as they arise.

Tableau's native functionality accommodates users of all technical backgrounds: just a few clicks reveal new insights, turning complex performance metrics into actionable intelligence for the entire Williams Racing team.

CONCLUSIONS

Three core conclusions emerge from our analysis of Williams Racing's 2015-2019 performance:

1. **High Variability in Race Outcomes**

Williams exhibited greater swings in grid-to-finish positions than other midfield rivals. While the team occasionally gained positions, inconsistency across different race weekends highlighted a vulnerability to changing conditions, setup decisions, and strategic decisions.

2. **Balanced Sectors Remain a Key Limitation**

Contrary to expectations, the largest qualifying deficits were observed not in technical, low-speed sectors, but in balanced sectors—those featuring mixed layouts of short straights and medium-speed corners. This suggests that all-around car performance, rather than low-speed cornering capabilities alone, is a primary constraint in qualifying pace.

3. **Rookies Proven to be Less Consistent than Experienced Teammates**

Statistical testing revealed that rookies exhibited higher lap-time variance than veterans during race sessions, indicating that experience does affect execution stability. This finding justifies targeted rookie development and support.

RECOMMENDATIONS

1. **Implement Live Sector-Tagging and Delta Tracking**

Introduce a system to tag sector types (technical, balanced, power) and monitor live time deltas during sessions. This enables more responsive and targeted race-weekend setup adjustments, ensuring resources are deployed where time is typically lost rather than relying on post-session outcomes.

2. **Prioritise Stability in Balanced Sectors Through Setup Trials**

Focus on setup trials and car development that improve predictability and control in balanced sectors. Evidence suggests that tackling these mixed segments—often featuring a combination of medium-speed turns and short straights—will yield the largest gains in qualifying pace.

3. **Focus on Rookie Consistency Training**

Rookie race consistency trails significantly behind their more experienced counterparts. Simulator work and race weekend practice should prioritise 'delivering the same lap over and over' instead of always chasing ultimate pace. Rewarding and monitoring progress in race consistency will accelerate the development curve and reliability of all new Williams drivers.

These recommendations are designed to reduce race-to-race underperformance and result variability, close sector-specific gaps and create a sustainable foundation for rookie development. Implementing them will help Williams turn historical insights into practical trackside lessons for future seasons.

DATA: SAMPLING

The analysis covers the **2015-2019 FIA Formula One World Championships**, a period marked by rapid technical changes and evolving competition. To ensure meaningful and comparable insights, we focused on ten Grand Prix circuits that remained fixtures on the calendar throughout these years:

- Monza (Italy)
- Monaco
- Silverstone (UK)
- Spa (Belgium)
- Barcelona (Spain)
- Singapore
- Interlagos (Brazil)
- Hungaroring (Hungary)
- Red Bull Ring (Austria)
- Suzuka (Japan)

This selection provides a stable reference for benchmarking, reduces the impact of rare or one-off tracks, and captures a well-rounded mix of power, technical, and balanced circuits. Analysing the same tracks over multiple years allowed for reliable trend analysis in driver consistency, qualifying performance, and race position changes.

Data was extracted from two sources:

1. **Structured** – The Kaggle [‘Formula 1 World Championship \(1950-2024\)’](#) dataset, containing historical race, constructor, and driver-level information.
2. **Unstructured** – [Fast F1 API](#), capturing granular lap times, sector splits, and telemetry, giving in-depth visibility into race weekend performance.

The project excludes data collected past 2020, where races were cancelled due to COVID-19, e.g. Singapore, Japan. These circuits later returned to the F1 calendar from 2022 onwards, but due to the potential time gap in the analysis, they were excluded from the project scope.

DATA: LIMITATIONS

While every effort was made to ensure a robust dataset, some limitations remain. The finest level of telemetry – such as lap-by-lap times and sector analysis – is only available from the 2018 season onward. As a result, more detailed qualifying comparisons (Strategic Question 2) are restricted to 2018 and 2019 data.

In some cases, sample sizes for certain KPIs or hypothesis tests fell below the ideal threshold for robust statistical analysis, especially when focusing on specific driver segments or rare events. The data for some questions also showed skew and irregular patterns, which can challenge traditional statistical methods. To mitigate this, non-parametric tests were used, but some uncertainty remains in interpreting these results.

For the KPI 3 Add-On Dashboard (focusing on driver consistency), the scope was broadened to include all races from 2015 to 2019 to better capture variability across different track types and race conditions. While this enhances completeness, it also introduces more diversity into the dataset, impacting comparability.

These limitations are typical in F1 analytics, given how real-world racing is shaped by engineering changes, unpredictable weather, and race incidents. By highlighting these constraints, we ensure conclusions are interpreted with the right caution and context – and note that, as data coverage improves, future analyses will become even more precise.

REFLECTIONS

Concluding this analysis highlighted the unique challenges and unexpected discoveries of robust statistical testing in motorsport data. Many core hypotheses used were structured around common perceptions held within the team, such as the presumption that rookie drivers struggle for race consistency versus veterans.

Every hypothesis test except one ([KPI 3 – ‘Driver Consistency’](#)) resulted in a null finding, showing the data did not support widely held beliefs. This outcome is significant in itself; disproving a commonly held assumption refocuses strategy and avoids bias. It also underscores the importance of letting statistical evidence, not expectation, drive insight – a quality frequently iterated upon in the project’s storytelling and design.

Applying rigorous hypothesis tests, even when the null is not rejected, demonstrates maturity in analytics: proving something right is just as valuable as identifying areas for improvement. The emphasis of the project remained on providing statistically backed recommendations.

Time management was another notable personal point for reflection. Keeping up with a complex analysis and ensuring regular progress tested personal discipline and project planning skills. Meeting key milestones with only minor delays was a rewarding outcome of applying consistent effort and flexibility – vital personal qualities for delivering robust work in fast-paced, deadline-driven environments.

FUTURE WORK

Data Filtering Methodology: The original KPI 3 analysis included all recorded lap times, including opening laps and pit stops, artificially widening our variability metrics due to their atypical pace and racing context. Initially, a static filter removing lap times over 120 seconds was applied.

To improve data quality, a dynamic filtering approach modelled after Formula 1’s “107% rule.” Was used. Instead of a fixed threshold, this method calculates the median lap time for each session and excludes outliers exceeding 105% of this median. This approach produced cleaner metrics by adapting to the specific conditions of each race session.

Future iterations could benefit from experimenting with different percentage thresholds to optimise the balance between removing noise and retaining valid edge-case data.

Statistical Sampling Considerations: A critical lesson emerged regarding data segmentation. When analysing specific subgroups (e.g. power vs technical by experience level), our sampling often became too narrow. Combined with feature engineering, this resulted in sample sizes below 30 and multimodal distributions violating parametric test assumptions.

This required non-parametric methods (e.g. Mann-Whitney U), which are more robust but less powerful for detecting subtle effects. Going forward, broader sampling strategies or developing hierarchical grouping approaches could help maintain adequate statistical power while preserving meaningful analysis results.

